

Exercise 1

The exercise is to investigate stability properties of the origin for the nonlinear system

$$x(k+1) = \underbrace{\begin{pmatrix} x_1(k) - x_1(k)^3 \\ -x_2(k)^3 - x_1(k)^3 \end{pmatrix}}_{:=g(x)}, \quad x(0) = x_0. \quad (1)$$

Determine a compact set $0 \subset \Omega \subset \mathbb{R}^2$ such that the origin $x_s = 0$ is a locally asymptotically stable equilibrium point for system (1) for all $x \in \Omega$.

Solution:

To conclude about asymptotic convergence to the origin for initial conditions in a compact set, we use the Lyapunov **direct** method¹.

Therefore, we need to find a (non-global) Lyapunov function that satisfies the following conditions (Lec. 6, Slide 30):

1. $V(0) = 0$ and $V(x) > 0, \forall x \in \Omega \setminus \{0\}$.
2. $V(g(x(k))) - V(x(k)) \leq -\alpha(x), \forall x \in \Omega$, with $\alpha(x) : \mathbb{R}^2 \rightarrow \mathbb{R}$ continuous and positive definite.
3. Ω is a positive invariant set for system (1) containing the origin.

We choose the following Lyapunov function

$$V(x) = x^T P x \text{ with } P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (2)$$

The first condition is obviously verified. For the second condition, we need to investigate

$$\begin{aligned} V(g(x(k))) - V(x(k)) &= V(x(k+1)) - V(x(k)), \\ &= (x_1 - x_1^3)^2 + (-x_2^3 - x_1^3)^2 - x_1^2 - x_2^2, && \text{(insert definition)} \\ &= x_1^2 - 2x_1^4 + x_1^6 + x_2^6 + 2x_2^3x_1^3 + x_1^6 - x_1^2 - x_2^2, && \text{(expand)} \\ &= 2x_1^4(x_1^2 - 1) + x_2^2(2x_2x_1^3 + x_2^4 - 1). && \text{(factor out positive terms)} \end{aligned} \quad (3)$$

In view of the third condition, we choose the set Ω as a sublevel set of the Lyapunov candidate $V(x)$, i.e., the ball with radius $\epsilon > 0$:

$$\mathcal{B}(\epsilon) := \{x \in \mathbb{R}^2 \mid \|x\|_2 \leq \epsilon\}.$$

Within $\mathcal{B}(\epsilon)$ we have that $|x_1| \leq \epsilon$ and $|x_2| \leq \epsilon$. Thus, for the second condition, we can upper-bound

$$2x_1^4(x_1^2 - 1) + x_2^2(2x_2x_1^3 + x_2^4 - 1) \leq 2x_1^4(\epsilon^2 - 1) + x_2^2(3\epsilon^4 - 1) \quad (4)$$

To ensure the right-hand side is a negative definite function (i.e., $-\alpha(x)$), we require

$$\epsilon^2 - 1 < 0 \Rightarrow \epsilon < 1,$$

$$3\epsilon^4 - 1 < 0 \Rightarrow \epsilon < \left(\frac{1}{3}\right)^{1/4} \approx 0.75$$

¹Concluding about local stability using, e.g., Lyapunov indirect method (Lec. 1c, Slide 51) generally does not help to determine a compact region of attraction $\Omega \supset \{0\}$.

This is satisfied by, e.g., $\epsilon = 1/2$, which leads to

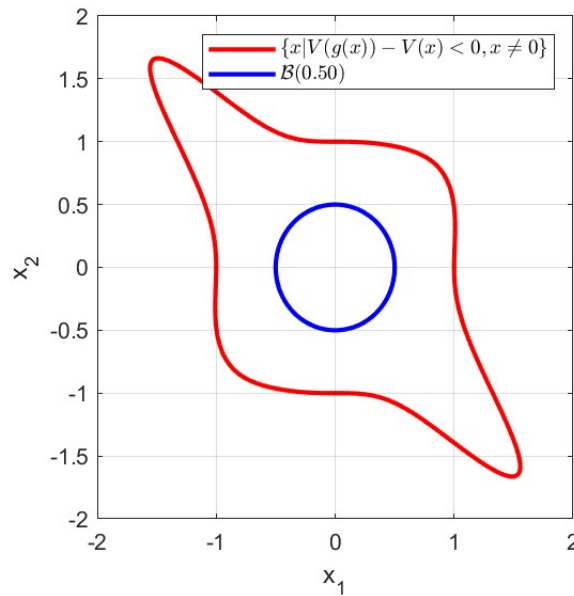
$$V(g(x(k))) - V(x(k)) \leq - \underbrace{\left(\frac{3}{2}x_1^4 + \frac{13}{16}x_2^2 \right)}_{\alpha(x)}$$

for all $x \in \mathcal{B}(0.5)$. Lastly, from the definition of the 2-norm we have

$$\mathcal{B}(\epsilon) = \{x \in \mathbb{R}^2 \mid \|x\|_2 \leq \epsilon\} = \{x \in \mathbb{R}^2 \mid x^\top x \leq \epsilon^2\} = \{x \in \mathbb{R}^2 \mid V(x) \leq \epsilon^2\}.$$

Thus, $\mathcal{B}(\epsilon)$ is an invariant set for system (1) (see Lec. 5, Slide 70). It follows that the origin $x_s = 0$ is a locally asymptotically stable equilibrium for system (1) within the compact set $\mathcal{B}(0.5) \subset \mathbb{R}^2$.

The plot shows the ball $\mathcal{B}(0.5)$ and the contour where $V(g(x)) - V(x) \leq 0$.



Exercise 2

Consider the following MPC problem

$$\begin{aligned} \min_{u_i} \quad & \alpha x_N^2 + \sum_{i=0}^{N-1} x_i^2 + u_i^2 \\ \text{s.t.} \quad & x_{i+1} = \frac{3}{2}x_i + u_i, \quad i = 0, \dots, N-1 \\ & x_0 = x(k) \end{aligned}$$

with $\alpha \in \mathbb{R}$, and $N = 1$. Is the origin an asymptotically stable equilibrium point for the closed-loop system under the resulting MPC control law for:

- i) $\alpha = \frac{1}{2}$?
- ii) $\alpha = \frac{3}{4}$?

Solution:

Strategy: Solve MPC problem analytically and investigate resulting closed-loop dynamics.

- Eliminate states (leave initial condition as parameter) by inserting the dynamics into the cost:

$$\min_{u_0} x_0^2 + u_0^2 + \alpha x_1^2 = \min_{u_0} x_0^2 + u_0^2 + \alpha \left(\frac{3}{2}x_0 + u_0 \right)^2$$

- Calculate minimum by setting the derivative of the cost w.r.t. u_0 to zero:

$$0 = \frac{\partial}{\partial u_0} \left(x_0^2 + u_0^2 + \alpha \left(\frac{3}{2}x_0 + u_0 \right)^2 \right)$$

$$0 = 2u_0 + 2\alpha \left(\frac{3}{2}x_0 + u_0 \right)$$

$$0 = 3\alpha x_0 + 2u_0(1 + \alpha)$$

$$u_0 = -\frac{3\alpha}{2(1 + \alpha)}x_0$$

- Obtain MPC control law:

$$u(k) = u_0(x(k)) = -\frac{3\alpha}{2(1 + \alpha)}x(k)$$

- Insert MPC control law into the dynamics:

$$x(k+1) = \frac{3}{2}x(k) - \frac{3\alpha}{2(1 + \alpha)}x(k)$$

$$= \frac{3}{2} \left(1 - \frac{\alpha}{1 + \alpha} \right) x(k)$$

$$= \begin{cases} \text{i) } x(k), & \text{for } \alpha = \frac{1}{2} \\ \text{ii) } \frac{6}{7}x(k), & \text{for } \alpha = \frac{3}{4} \end{cases}$$

- Conclusions:

- Origin is not asymptotically stable (just stable). Any subset of $\mathbb{R}$ is a positive invariant set for the system under the given MPC controller.
- Origin is asymptotically stable.

- Observations:

- A larger terminal cost leads to stability. Why don't we just choose α 'very' large? In the limit case $\alpha \gg 1$, the stage cost $x^2 + u^2$ will be neglected by the MPC plus numerics get difficult.